

Review of MPPT Techniques for Dynamic Partial Shading in Solar PV Systems

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Abstract—Dynamic Partial Shading Conditions (DPSC) significantly impact the energy output of solar photovoltaic systems, leading to multiple local maxima in the power-voltage (P-V) curve and reducing energy efficiency. To resolve this challenge, advanced Maximum Power Point Tracking techniques have been developed to ensure optimal power extraction. This review focuses on three cutting-edge MPPT—Incremental Conductance with Differential Optimization (INC-DFO), Sovereign Butterfly Optimization (SBO), and Whale Optimization with Differential Evolution (WODE)—which have shown promising results in handling rapid shading transitions. These techniques are evaluated based on key performance parameters, including tracking speed, accuracy, adaptability to shading patterns, robustness, computational complexity, and convergence behavior. The analysis highlights that while INC-DFO offers moderate tracking speed and reliability, SBO provides the fastest and most robust convergence. WODE achieves a balance between precision and computational efficiency, making it an ideal choice for real-time applications. This review serves as a critical resource for researchers and engineers seeking efficient MPPT strategies to enhance PV system performance under dynamic shading scenarios.

Index Terms—Dynamic Partial Shading Conditions (DPSC), Maximum Power Point Tracking (MPPT), Photovoltaic (PV) Systems, Renewable Energy.

I. INTRODUCTION

Nearly all nations across the globe are embracing solar photovoltaic technologies as a substitute for traditional energy production, owing to numerous benefits such as decreased greenhouse gas emissions, the limitless availability of solar power, and environmentally friendly qualities, among others. India is likewise aiming for an electrical output of 100 GW by 2022 through various solar parks, both large and small, to satisfy the increasing energy requirements [1].

Literature presents numerous MPPT algorithms established toward optimize power extraction from solar PV methods also enhance entire method performance [2]. The traditional algorithm operates effectively under standard temperature and irradiance conditions, successfully achieving maximum power. However, in reality, environmental conditions are not static; both temperature and irradiation are subject to fluctuations [3], [4]. Under scenarios where irradiation levels change swiftly,

standard MPPT methods struggle to efficiently locate the MPP within a brief timeframe. As a result, the performance and operational efficiency of solar PV systems significantly decline [5], [6]. Moreover, partial shading conditions caused by varying solar irradiation incidents across different panels indications toward the appearance of several highest points on PV typicals.

Under conditions of partial shadowing, multiple maximum points (local MPPs) and one global MPP can be identified. The standard algorithm doesn't distinguish among local as well as global MPPs [7], [8]. Consequently, the traditional algorithm initiates its monitoring from the nearby MPP, treating it as the global MPP, which consequently leads to a decline in the method's overall presentation as well as efficiency. In order to tackle the aforementioned challenges, researchers worldwide have created various advanced MPPT techniques that have achieved limited success in addressing the problem. Nonetheless, there remains a need for research focused on enabling more rapid tracking of the MPP with enhanced precision. Generally speaking, MPPT algorithm of conventional can be divided into 2 categorizes: Online techniques as well as offline, based on requisite constraints as well as values. The offline technique essentially relies on a model-driven approach. Typically, Control signals are typically derived from the instantaneous values of PV output voltage (or current). Various traditional MPPT strategies have been documented in the literature each possess unique advantages and disadvantages, making it complicated to choose one from the numerous strategies available. Current review literature assesses different MPPT algorithms; nonetheless, they can sometimes be challenging to comprehend [9], [10].

The operation of PV array is influenced via changes in irradiance from one panel toward another caused by the shadows created by transient trees, clouds near structures, alterations in weather, accumulation of dirt, degradation impact of panel, and more changing with time. Consequently, Partial Shading Conditions (PSC) and Dynamic Partial Shading Conditions (DPSC) is able to arise in PV system because of any these factors. When PSC occurs in PV method, the Voltage–Power (V-P) curve may exhibit either one peak otherwise several peaks. The presence of a one peak or multiple peaks on V-P curves is contingent upon the intensity of sunlight irradiating

the PV system panels. Should every PV panels experience uniform irradiance, there will be an individual P-V curve on peak. In contrast, uncertainty single, two, otherwise every of the PV panels encounter shadowing effects, multiple peaks will appear. Consequently, the operating point that satisfies the MPP requirement varies according to the environmental factors. Toward improve power production of PV systems, MPPT method, it serves as a control strategy capable of continuously monitoring the MPP during functioning, is essential. However, there remains an opportunity for research to examine the impact of the photovoltaic system concerns shades under dynamic PSC. Here, we examine the effects of the MPPT in relation towards the dynamic PSC.

- A review of MPPT techniques under dynamic partial shade conditions (PSC), highlighting their strengths and limitations.
- Analysis of the impact of dynamic partial shading conditions on PV system performance.
- Identification of research gaps and future directions for developing more efficient MPPT techniques capable of rapid and accurate tracking under dynamic partial shading conditions.

The remainder of this paper is organized as : Section 2 covers the solar PV systems. Section 3 discusses various MPPT techniques and provides a comparative analysis. Finally, Section 4 summarizes the key findings and future research directions.

II. SOLAR PHOTOVOLTAIC SYSTEM

Solar power has become the leading form of renewable energy owing toward its usability in any environment and its daily availability. Thanks to its straightforward design, minimal environmental impact, negligible or nonexistent carbon emissions, and low upkeep expenses, it has established itself as a key renewable resource. Currently, solar energy stands as the global favorite among alternative energy sources. However, solar PV system faces the inevitable challenge of current as well as voltage nonlinearity, which can arise in both typical scenarios and PSC. Generally, PV systems contain a distinct operating point at which power output reaches its peak. Therefore, it is essential for PV system toward function now to achieve optimal efficacy. To enhance power output in all typical circumstances, the PV system utilizes the MPPT mechanism [11].

A. PV system

Generally, the features of photovoltaic (PV) panels remain illustrated through I-V plus P-V graphs, in Figure 1. The situation remains important to highlight that there exists a particular operational point where the power o/p reaches its peak. This distinct point is known as the MPP also indicated via acronym MPP. Due to the inherent PV panels low efficacy, PV systems need to consistently function near the MPP to capture the maximum possible energy. A PV cell represents a basic component that makes use of PV result toward transform solar power into electricity energy. Upon receiving sunshine,

a cell of PV produces electrical energy. The analogous circuit of standard PV cell comprises sequence as well as parallel resistances. The single diode typical represents greatest precise among various PV cell modeling approaches.

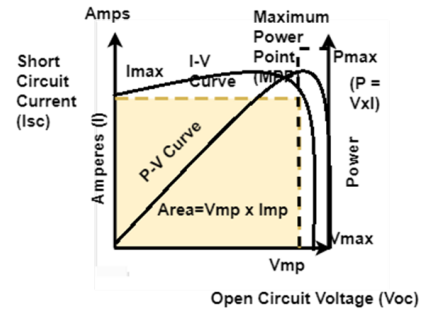


Fig. 1. PV-IV characteristics

Though, efficiency of PV module can readily remain influenced via weather circumstances. The current in short-circuit is linearly related toward the degree of solar radiation, while open circuit voltage is considerably impacted via the cell temperature. The I-V plus P-V graphs aimed at varying temperatures.

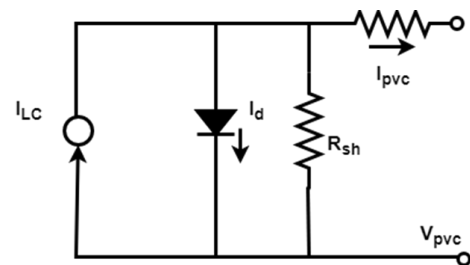


Fig. 2. PV cell method

It is important to recognize that there exists a designated point of operation where the o/p power reaches its peak. This distinct point is known as the MPP, represented via the abbreviation MPP in I-V as well as P-V graphs. Considering how inefficient photovoltaic panels are, photovoltaic systems need to function consistently near the MPP to capture the highest possible amount of energy [12].

The objective of the MPPT method is to identify the maximum power output of PV panel. By various fluctuating temperatures as well as solar radiation levels, MPP also shifts. Therefore, in order to continuously adjust the MPP as an operational point across a wide spectrum of i/p (including temperature as well as solar irradiance), MPPT technique is required. The implementation of MPPT functions as a DC/DC converter positioned among load, which is regulated via photovoltaic modules and the tracking algorithm. The tracking algorithm essentially acquires inputs such as PV module voltage as well as current, and coordinates by duty cycles of DC/DC converter toward establish operational point of system at MPP [13]. Nevertheless, there exist additional

significant factors to take into account when developing MPPT strategies for photovoltaic systems, such as efficacy, method of implementation, complexity, lost energy, and expenses. In light of these considerations, multiple MPPT methods have been devised and categorized into three primary classifications, specifically dynamic partial shading situations a wide-ranging categorization of MPPT strategies.

III. MPPT SCHEMES FOR DYNAMIC PARTIAL SHADING

Partial shading during variable conditions diminishes the total PV system performance. To draw out maximum amount of power aimed at PV system in this uneven ecological situation, it is crucial to implement partial shading alleviation techniques for PV system operating in either grid-connected otherwise standalone configurations. Traditional MPPT techniques utilize fixed step intervals as well as often get stuck at local maxima, thus they are incapable of effectively tracking the MPP in partial shading scenarios. To counteract these issues arising from standard techniques in fluctuating conditions, metaheuristic strategies like SBO and WODE have been suggested as MPPT methodologies [13-14].

A. Hybrid MPPT Technique

The InC-DFO method denotes a computational approach that combines the strengths of both algorithms. These binary techniques are used in sequence to improve algorithm efficiency. The output generated by the InC technique, using a variable step size sets the beginning for the DFO method. Nevertheless, the InC-DFO technique is utilized across all weather scenarios. The InC-DFO algorithm functions in two phases: during the initial phase, the InC technique with a variable step size is utilized to ensure optimal performance under uniform irradiation conditions as well as shorter convergence and settling times. However, this method tends to get stuck at local maxima (LM), leading to power loss, as illustrated in Figure 3[15]

To tackle these challenges, the InC algorithm utilizes a dynamically adjusted step size, which is expressed as:

$$d_q = d_{q-1} \pm N \cdot \left| \frac{\text{dev} \times p}{\text{dev} \times v - \text{dev} \times i} \right| \quad (1)$$

In this expression, d_{q-1} and q_k represent the step size values of the boost converter at the previous time step ($q - 1$) and the current time step (q), respectively.

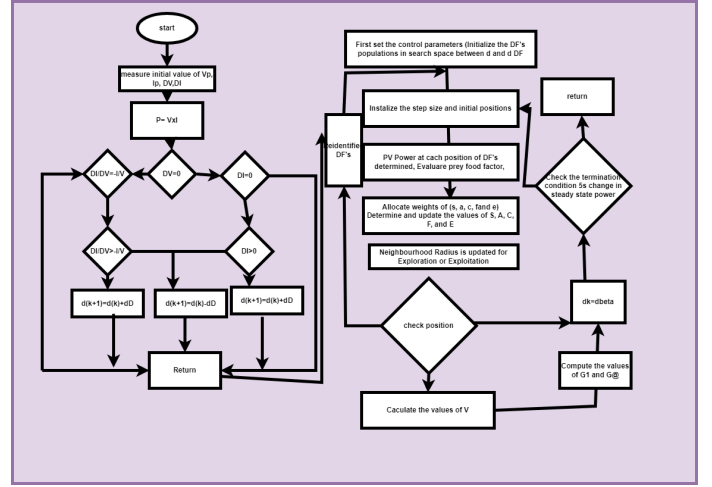


Fig. 3. INC-DFO

B. Sovereign Butterfly Optimization

For sophisticated MPPT, this document introduces an improved method for grid-tied photovoltaic (PV) systems utilizing a flyback converter alongside the Sovereign Butterfly Optimization technique. The fitness function, indicating the population span of maximum power, is articulated as the proportion of each power output to the peak power, as illustrated in Figure 4 [16]

$$F_i = \frac{P_i}{P_{\max}} \quad (2)$$

The migration procedure consists of revising the duty cycle parameters according to the Sovereign Butterfly Optimization method, as illustrated in Equations (2) through (6).

$$X_{j,m}^{(t+1)} = X_{s1,M} \quad (3)$$

A random digit is chosen by applying the next formula. In Formula (3), i represents the transfer duration in a different scenario; provided that $r > p$, then K , the component of the newly formed generation's position, is determined utilizing the following equation.

$$r = \text{rand} \times p \quad (4)$$

In Expression (4), $X_{t,s2:M}$ signifies the K -th component of $X_{t,s2:M}$ at the t -generation of duty cycle parameter $r2$; a proportion of duty cycle inland 1 is indicated by the symbol P , which represents the count of local optimum values that are revised through an updating value.

$$X_{j,m}^{(t+1)} = X_{s2,M} \quad (5)$$

For every component of the child's duty cycle j ; the revised formula is presented as illustrated underneath. In Formula (5), $X_{j,k}^{(t+1)}$ represents the K -th component that is articulated as the butterfly j position, whereas $X_{\text{best},M}$ signifies the K -th component of X_{best} during the prevailing generation t in lands

1 and 2. In this scenario, if $\text{rand} > P$, it is modified according to the subsequent equation.

$$X_{j,k}^{(t+1)} = X_{\text{best},M} \quad (6)$$

In Formula (6), if rand exceeds the acquired fitness value, then an updated position has been established. The system persistently monitors the duty cycle (DCp) as the reference point (Rp), guaranteeing effective MPPT. To commence the global MPPT tracking procedure, the power change ratio (Pbcp) is computed.

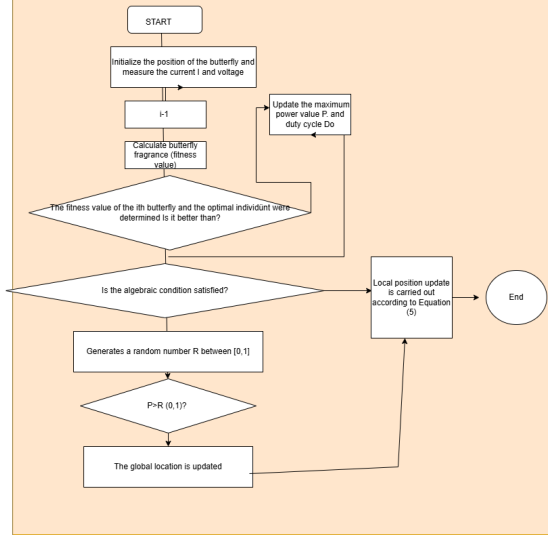


Fig. 4. SBO

C. WODE

The WO approach derives from the foraging technique of a humpback whale. This foraging style centers around a bubble-net feeding system characterized by a contracting circular spiral movement. From their starting point, humpback whales begin their search in a random manner (relative to the starting position). Subsequently, the WOA compels the search to occur on a more global scale, utilizing a coefficient vector of random (A). When $|A| > 1$, the humpback whales commence exploration throughout the whole area as shown in Figure 18. This phenomenon is mathematically articulated in [17].

$$D_{ij}(G+1) = D_{\text{rand}} - A \cdot d_{ij} \quad (7)$$

In this phase, the humpback whale identifies the target and begins to circle around it. While circling, the whale adjusts its location in relation to the optimal prey it has encountered. This movement occurs when $|A| < 1$. The representation of this is as follows,

$$d_{ij} = |C \cdot D_{\text{best}}(G) - D_{ij}(G)| \quad (8)$$

$$D_{ij}(G+1) = D_{\text{best}} - A \cdot d_{ij} \quad (9)$$

Where, $D_{\text{best}}(G)$ refers to the optimal D following the G -th Step.

In the hunting strategy of the bubble net, the movement of the whale is segmented into two sections with equal 50-50% likelihood: straight-line movement along the diminishing circle as well as rotational movement along the helical trajectory. This can be expressed as,

$$d'_{ij} = |D_{\text{best}}(G) - D_{ij}(G)| \quad (10)$$

The DE mutation mechanism produces a trial vector obtained from the parent vector through the use of differential coefficients weighted by a scale factor for mutation ξ . This technique is expressed mathematically by,

$$U_i(G) = D_{i1}(G) + \xi \cdot (D_{i2}(G) - D_{i3}(G)) \quad (11)$$

Here ‘ i ’ represents the existing group count. DE chooses the optimal choice $D_i(G+1)$ for the subsequent generation, selecting both the parent and offspring according to the highest value of the fitness function f .

IV. COMPARATIVE ANALYSIS OF DYNAMIC PARTIAL SHADING TECHNIQUES

In the context of dynamic partial shading (DPS) conditions:

- **INC-DFO (Incremental Conductance with Differential Evolution Optimization):** This method combines the well-known Incremental Conductance (INC) technique with an optimization approach. It is specifically designed for steady-state tracking even under dynamic shading conditions.
- **SBO (Sovereign Butterfly Optimization):** This technique employs the principles of butterfly-inspired mechanisms, providing fast and accurate MPP tracking in various environmental conditions. SBO stands out due to its adaptability and ability to handle the rapid fluctuations associated with dynamic partial shading scenarios, ensuring that MPP tracking remains precise.
- **WODE (Whale Optimization with Differential Evolution):** The WODE method integrates Whale Optimization with Differential Evolution, combining global search skills of whale optimization with the fine-tuning capabilities of differential evolution. This combination offers an efficient solution for MPP tracking, even in highly volatile environments where partial shading can be complex and rapidly changing.

TABLE I
ANALYSIS OF DYNAMIC PARTIAL SHADING TECHNIQUES

Parameter	Tracking Speed	Accuracy	Adaptability (PSCs)	Robustness	Computational Complexity	Convergence Behavior
INC-DFO	Moderate	High	Moderate	Moderate	Moderate	Reliable but slower
SBO	High	High	High	High	High	Fast and Reliable
WODE	Moderate-High	Very High	Very High	High	High	Balanced

Each of these methods brings its unique strengths to dynamic partial shading situations, balancing speed, accuracy, and adaptability to ensure that optimum power is harvested from the PV system despite fluctuating shading conditions.

Table II summarizes the key MPPT techniques under dynamic conditions. Each method is briefly characterized by its core mechanism and effectiveness in handling rapid environmental variations.

TABLE II
MPPT TECHNIQUES FOR DYNAMIC CONDITIONS

Type	Method	Brief Description
Dynamic	INC-DFO	Combines Incremental Conductance with metaheuristics for improved steady-state tracking.
	SBO	Employs butterfly-inspired strategy for fast and accurate MPP tracking in all environments.
	WODE	Hybrid of Whale Optimization and Differential Evolution for robust tracking under volatile conditions.

V. SUPERIOR PERFORMANCE OF DYNAMIC SHADING CONDITION MPPT TECHNIQUES: A CASE ANALYSIS.

Dynamic MPPT techniques leverage advanced tracking strategies to accurately identify and maintain the Maximum Power Point (MPP) even under fast-varying environmental conditions. Compared to traditional methods, dynamic MPPT approaches exhibit higher tracking efficiency, reduced oscillations around the MPP, and faster response rates. This ensures improved energy extraction and enhanced system stability.

A. Representative Case: Performance Under Fast-Varying Environmental Conditions

TABLE III
COMPARISON OF TRACKING EFFICIENCY, OSCILLATION NEAR MPP, RESPONSE RATE, AND TRACKING TIME FOR VARIOUS MPPT TECHNIQUES.

Technique	Eff. (%)	Osc. (%)	Rate (%)	Time (s)
IC	95.94	20.54	96.27	0.755
PSO	97.71	2.00	97.23	0.795
SBO	99.30	2.00	98.50	0.660

PSO - Particle Swarm Optimization; SBO - Sovereign Butterfly Optimization; IC - Incremental Conductance.

The comparative analysis reveals that these techniques excel in minimizing power losses caused by rapid changes in solar irradiance or temperature. By maintaining a stable and precise tracking performance, they ensure reliable operation of photovoltaic systems under highly varying atmospheric conditions as shown by Table III.

B. Representative Case: Performance Under Dynamic partial Shading Conditions

Dynamic Maximum Power Point Tracking (MPPT) techniques, particularly Sovereign Butterfly Optimization (SBO), exhibit superior performance over conventional MPPT algorithms such as Differential Particle Swarm Optimization (DPSO) and Incremental Conductance (IC) under dynamic partial shading conditions. These algorithms were tested across

two different patterns with varying irradiance levels and time intervals (0-10s, 10-20s, and 20-30s) as shown in table IV.

TABLE IV
PERFORMANCE COMPARISON OF ALGORITHMS ACROSS PATTERNS

Time Interval	Pattern-1 (Pmpp = 2430.8 W)	Pattern-2 (Pmpp = 2264.2 W)
0-10s (800W/m ² - 1000W/m ²)	DPSO: 4.8s IC: 0.3s SBO: 0.27s	DPSO: 2.85s IC: 0.5s SBO: 0.42s
10-20s (1000W/m ² - 500W/m ²)	DPSO: 0.5s IC: 0.4s SBO: 0.35s	DPSO: 0.8s IC: 0.5s SBO: 0.4s
20-30s (500W/m ² - 800W/m ²)	DPSO: 0.4s IC: 0.4s SBO: 0.3s	DPSO: 0.6s IC: 0.5s SBO: 0.45s

DPSO - Differential Particle Swarm Optimization; SBO - Sovereign Butterfly Optimization; IC - Incremental Conductance.

VI. CONCLUSION

This paper presented a comparative analysis of MPPT techniques under dynamic partial shading conditions (DPSC), emphasizing the challenges posed by rapid irradiance variations in PV systems. Advanced optimization methods such as Sovereign Butterfly Optimization (SBO) and the hybrid Whale Optimization with Differential Evolution (WODE) outperformed traditional techniques, offering fast convergence, high accuracy, and robustness. These methods are particularly suited for real-world DPSC scenarios, ensuring reliable and efficient power extraction. The findings highlight the necessity of adopting intelligent optimization algorithms for effective MPPT under dynamic conditions. Future work will focus on real-time hardware implementation to validate and further enhance these techniques for practical PV deployments.

While significant progress has been made, several challenges and opportunities for future research remain. Future directions include: (i) real-time validation of SBO/WODE, (ii) computational optimization for embedded platforms, and (iii) cost-effective deployment in smart grids.

- **Integration with emerging technologies:** Future investigations should explore the integration of dynamic MPPT techniques, such as SBO, with advanced grid-forming inverters and hybrid power systems incorporating electric vehicles and thermal storage.
- **Enhanced computational efficiency:** The computational complexity of soft computing and hybrid techniques necessitates further research to optimize algorithms for real-time implementation with minimal processing resources.
- **Economic feasibility and scalability:** Future studies should focus on minimizing the costs associated with MPPT systems, including sensor requirements and implementation, to facilitate widespread adoption in residential and commercial PV systems.
- **Comprehensive field validation:** Extensive experimental validation under diverse and realistic shading patterns is essential to confirm the efficacy and reliability of advanced MPPT techniques in large-scale PV installations.

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